

Cybersecurity Self-Study Notes Kit — 2026

How to learn, question yourself, check knowledge, and a full domain cheat sheet. Print-friendly.

1. How to take notes that stick

The Cyber Cornell Sheet: For every topic, make 3 columns — Concept / Command / Why it matters. Left: term. Middle: exact command / example. Right: what breaks if you get this wrong, and how an attacker abuses it.

- Explain in plain language first, then add the technical term. If you cannot explain to a friend, you do not know it yet.
- One note = one attack/defense idea. Small cards beat long pages.
- Always add: What is it? How does it work? How do I break it? How do I detect/fix it?
- Link notes: Network → Protocol → Vulnerability → Tool → Log signature.
- Keep a Lab Log: date, what you tried, what worked, screenshot/command, what you learned.

2. Self-questioning system

Feynman 4-questions, ask for every topic:

- 1. Can I explain this in 2 simple sentences?
- 2. What is a real example I have seen in a lab?
- 3. How would an attacker use this? How would a defender catch it?
- 4. What command/log would prove it?

Socratic prompts to keep in your notebook: What is the trust boundary here? Where is the input? What if that input is malicious? Where would this show up in logs? What is the fix that does not break users?

3. How to check your knowledge

- Active recall: close notes, write everything you remember. 15 min daily.
- Spaced flashcards: Anki. Front = scenario, Back = tool + detection. 20 cards/day.
- Lab check: Can you do it without a walkthrough? TryHackMe / HackTheBox / PortSwigger / LetsDefend.
- Teach-back: record a 3-min voice note explaining the topic.
- Weekly quiz: 10 questions — 4 recall, 3 scenario, 3 commands.
- Score yourself: Green = can teach it, Yellow = needs notes, Red = redo lab.

4. Full Cybersecurity Domain Map — what to learn

Domain	Core topics to note
1. IT Foundations	Networking TCP/IP, DNS, HTTP, OSI. Linux/Windows CLI. Virtualization. Python/Bash basics.
2. Security Fundamentals	CIA triad, AAA, risk, threats/vulns, encryption basics, PKI, identity.
3. Network Security	Firewalls, IDS/IPS, VPN, segmentation, Wireshark, Nmap, Zeek/Suricata.
4. OS & Endpoint	Hardening, permissions, AV/EDR, Sysmon, autoruns, memory basics.
5. Web App Security	OWASP Top 10, HTTP, cookies/sessions, Burp Suite, SQLi/XSS/CSRF/IDOR.
6. Cryptography	Symmetric/AES, Asymmetric/RSA, hashes, TLS, certificates, common mistakes.
7. SOC / Blue Team	SIEM Splunk/ELK, log analysis, MITRE ATT&CK, incident triage, playbooks.
8. Offensive / Red Team	Recon, scanning, exploitation, post-exploitation, Metasploit, reporting.
9. Cloud Security	AWS/Azure/GCP IAM, S3/storage misconfig, containers, CSPM, cloud logs.
10. DFIR	Disk/memory forensics, timeline analysis, Volatility, Autopsy, chain of custody.
11. Malware Basics	Static/dynamic analysis, sandboxes, YARA rules, IOCs.
12. GRC & Compliance	ISO 27001, NIST CSF, risk assessments, policies, audit basics. India: DPDP Act, CERT-In.
13. DevSecOps	SAST/DAST, CI/CD security, secrets management, SBOM.
14. Wireless / Mobile / OT	WPA2/3, Android/iOS basics, SCADA/ICS awareness.

5. Cheat Sheets — copy these into your notebook

Networking Quick

TCP 3-way: SYN → SYN-ACK → ACK. Common ports: 22 SSH, 25 SMTP, 53 DNS, 80 HTTP, 443 HTTPS, 445 SMB, 3389 RDP. CIDR: /24 = 256 IPs, /16 = 65k. Check: `ip a / ping / traceroute / nslookup / netstat -tulpn`

Linux Quick

`ls -la, cd, cat, grep -Rin, find / -perm -4000 2>/dev/null, ps aux, netstat -tulpn, chmod 755, chown, useradd, journalctl -xe, tar -czvf`. Logs: `/var/log/auth.log, /var/log/syslog`

Nmap / Wireshark

`nmap -sV -sC -p- ; nmap -sU --top-ports 100 ; nmap --script vuln`. Wireshark filters: `http, ip.addr==x, tcp.port==443, dns, http.request.method==POST`

Web App — OWASP Top 10 prompts

- Broken Access Control — can I access other users' IDs? `/admin?`
- Crypto Failures — HTTP? weak ciphers? secrets in JS?
- Injection — ' OR 1=1 -- | alert(1)
- Insecure Design — is there rate limiting? password reset flow safe?
- Security Misconfig — default creds? directory listing? headers missing?
- Vuln Components — old libraries? check version strings
- Auth Failures — brute force? weak session? MFA bypass?
- Integrity Failures — unsigned updates? CI secrets leaked?
- Logging Failures — do failed logins get logged?
- SSRF — can I make the server fetch `http://169.254.169.254/` ?

SOC / SIEM — Triage cheat

- 1. What alerted? Rule name, severity, host/user
- 2. Is it true positive? Check process tree, network, auth logs
- 3. Scope: one host or many? lateral movement?
- 4. Contain: isolate host, disable account, block IOC
- 5. Eradicate/Recover: kill process, patch, restore
- 6. Close with timeline + MITRE technique ID
- KQL/SPL basics: `index=* failed | stats count by user | where count >5`

Crypto quick

Hash = integrity, not encryption. Never roll your own crypto. TLS 1.2+ only. Store passwords = Argon2/bcrypt + salt. AES-GCM > AES-CBC. RSA ≥2048.

Cloud IAM quick

Least privilege always. No `*:*` policies. Enable MFA, CloudTrail/Monitor logs, rotate keys, S3 buckets private, encrypt at rest, check public security groups.

Incident Response — 6 phases

1 Preparation 2 Identification 3 Containment 4 Eradication 5 Recovery 6 Lessons Learned

Reporting — 1-page finding template

Title / Severity / Affected asset / Description / Proof-of-Concept / Impact / Remediation / References (CVE/OWASP)

6. Your Note Template — print and reuse

Topic:	_____ Date: _____
Plain explanation:	

Key commands / tools:	
Attacker view:	Defender view:
Log signature / IOC:	
Self-test question:	Answer:
Confidence:	Green / Yellow / Red Next review: _____

7. 30-day knowledge-check cycle

- Days 1–10: Foundations — 2 flashcards per concept, 1 lab/day
- Days 11–20: Web + Network — 1 Burp lab + 1 Nmap/Wireshark exercise daily
- Days 21–30: SOC — 5 LetsDefend alerts/day, write 1 IR report/week
- Every Sunday: 30-min blank-page recall of the week, update Red/Yellow/Green list
- End of month: 1 portfolio write-up published to GitHub/LinkedIn

Study stack that works in 2026: TryHackMe Pre-Security, PortSwigger Web Academy, LetsDefend SOC, OverTheWire Bandit, CyberDefenders, HackTheBox Academy. Take Cornell notes, Anki recall, lab first.

Ethics: only test systems you own or have permission for. Learn to break to protect.